

REFERENCES

This page lists all references cited across the workshop materials.

- Bakka, Haakon, Håvard Rue, Geir-Arne Fuglstad, et al. 2018. "Spatial Modeling with R-INLA: A Review." *WIREs Computational Statistics* 10 (6): e1443. <https://doi.org/10.1002/wics.1443>.
- Bakka, Haakon, Jarno Vanhatalo, Janine B Illian, Daniel Simpson, and Håvard Rue. 2019. "Non-Stationary Gaussian Models with Physical Barriers." *Spatial Statistics* 29: 268–88.
- Blangiardo, Marta, and Michela Cameletti. 2015. *Spatial and Spatio-Temporal Bayesian Models with R-INLA*. John Wiley & Sons. <https://doi.org/10.1002/9781118950203>.
- Bryce, Erin, Luigi Lombardo, Cees Van Westen, Hakan Tanyas, and Daniela Castro-Camilo. 2022. "Unified Landslide Hazard Assessment Using Hurdle Models: A Case Study in the Island of Dominica." *Stochastic Environmental Research and Risk Assessment* 36 (8): 2071–84.
- Cameletti, Michela, Finn Lindgren, Daniel Simpson, and Håvard Rue. 2013. "Spatio-Temporal Modeling of Particulate Matter Concentration Through the SPDE Approach." *ASTA Advances in Statistical Analysis* 97 (2): 109–31.
- Fichera, Andrew, Rachel King, Jarrod Kath, David Cobon, and Kathryn Reardon-Smith. 2023. "Spatial Modelling of Agro-Ecologically Significant Grassland Species Using the INLA-SPDE Approach." *Scientific Reports* 13 (1): 4972.
- Goicoa, Tomás, Aritz Adin, Mariña Dolores Ugarte, and James S Hodges. 2018. "In Spatio-Temporal Disease Mapping Models, Identifiability Constraints Affect PQL and INLA Results." *Stochastic Environmental Research and Risk Assessment* 32 (3): 749–70.
- Gómez-Rubio, Virgilio. 2020. *Bayesian Inference with INLA*. Chapman & Hall/CRC. <https://doi.org/10.1201/9781315175584>.
- Krainski, Elias T., Virgilio Gómez-Rubio, Haakon Bakka, et al. 2019. *Advanced Spatial Modeling with Stochastic Partial Differential Equations Using R and INLA*. Chapman & Hall/CRC. <https://doi.org/10.1201/9780429031892>.
- Lamouroux, Jérémy, Alizée Geffroy, Sébastien Leblond, Caroline Meyer, and Isabelle Albert. 2025. "Addressing Spatial Confounding in Geostatistical Regression Models: An r-INLA Approach." *Methods in Ecology and Evolution* 16 (9): 2082–97.
- Laxton, Megan R, Óscar Rodríguez de Rivera, Andrea Soriano-Redondo, and Janine B Illian. 2023. "Balancing Structural Complexity with Ecological Insight in Spatio-Temporal Species Distribution Models." *Methods in Ecology and Evolution* 14 (1): 162–72.
- Lee, Sophie A, Theodoros Economou, and Rachel Lowe. 2022. "A Bayesian Modelling Framework to Quantify Multiple Sources of Spatial Variation for Disease Mapping." *Journal Of The Royal Society Interface* 19 (194): 20220440.

- Lindgren, Finn, David Bolin, and Håvard Rue. 2022. "The SPDE Approach for Gaussian and Non-Gaussian Fields: 10 Years and Still Running." *Spatial Statistics* 50: 100599.
- Lindgren, Finn, Johan Lindström, and Håvard Rue. 2010. *An Explicit Link Between Gaussian Fields and Gaussian Markov Random Fields; the Spde Approach*.
- Martino, Sara, Daniela Silvia Pace, Stefano Moro, et al. 2021. "Integration of Presence-Only Data from Several Sources: A Case Study on Dolphins' Spatial Distribution." *Ecography* 44 (10): 1533–43.
- Miller, David L, Richard Glennie, and Andrew E Seaton. 2020. "Understanding the Stochastic Partial Differential Equation Approach to Smoothing: DL Miller Et Al." *Journal of Agricultural, Biological and Environmental Statistics* 25 (1): 1–16.
- Moraga, Paula. 2019. *Geospatial Health Data: Modeling and Visualization with R-INLA and Shiny*. Chapman & Hall/CRC. <https://doi.org/10.1201/9780429341823>.
- Moraga, Paula, Christopher Dean, Joshua Inoue, Piotr Morawiecki, Shahzeb Raja Noureen, and Fengpei Wang. 2021. "Bayesian Spatial Modelling of Geostatistical Data Using INLA and SPDE Methods: A Case Study Predicting Malaria Risk in Mozambique." *Spatial and Spatio-Temporal Epidemiology* 39: 100440.
- Orozco-Acosta, Erick, Aritz Adin, and Mariña Dolores Ugarte. 2021. "Scalable Bayesian Modelling for Smoothing Disease Risks in Large Spatial Data Sets Using INLA." *Spatial Statistics* 41: 100496.
- Palmi-Perales, Francisco, Finn Lindgren, and Virgilio Gómez-Rubio. 2025. "Bayesian Inference for Case-Control Point Pattern Data in Spatial Epidemiology with the Inlabru r Package." *arXiv Preprint arXiv:2503.14954*.
- Riebler, Andrea, Sigrunn H. Sørbye, Daniel Simpson, and Håvard Rue. 2016. "An Intuitive Bayesian Spatial Model for Disease Mapping That Accounts for Scaling." *Statistical Methods in Medical Research* 25 (4): 1145–65. <https://doi.org/10.1177/0962280216660421>.
- Rivera, Óscar Rodríguez de, Juncal Espinosa, Javier Madrigal, Marta Blangiardo, and Antonio López-Quiñez. 2025. "Spatio-Temporal Marked Point Process Model to Understand Forest Fires in the Mediterranean Basin: Ór d. Rivera Et Al." *Journal of Agricultural, Biological and Environmental Statistics* 30 (3): 700–729.
- Rue, Håvard, Sara Martino, and Nicolas Chopin. 2009. "Approximate Bayesian Inference for Latent Gaussian Models by Using Integrated Nested Laplace Approximations." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 71 (2): 319–92. <https://doi.org/10.1111/j.1467-9868.2008.00700.x>.
- Rue, Håvard, Andrea Riebler, Sigrunn H. Sørbye, Janine B. Illian, Daniel P. Simpson, and Finn K. Lindgren. 2017. "Bayesian Computing with INLA: A Review." *Annual Review of Statistics and Its Application* 4: 395–421. <https://doi.org/10.1146/annurev-statistics-060116-054045>.
- Seaton, Fiona M, Susan G Jarvis, and Peter A Henrys. 2024. "Spatio-Temporal Data Integration for Species Distribution Modelling in r-INLA." *Methods in Ecology and Evolution* 15 (7): 1221–32.

- Simpson, Daniel, Janine Baerbel Illian, Finn Lindgren, Sigrunn H Sørbye, and Havard Rue. 2016. "Going Off Grid: Computationally Efficient Inference for Log-Gaussian Cox Processes." *Biometrika* 103 (1): 49–70.
- Simpson, Daniel, Håvard Rue, Andrea Riebler, Thiago G. Martins, and Sigrunn H. Sørbye. 2017. "Penalising Model Component Complexity: A Principled, Practical Approach to Constructing Priors." *Statistical Science* 32 (1): 1–28. <https://doi.org/10.1214/16-STS576>.
- Van Niekerk, Janet, Haakon Bakka, Håvard Rue, and Olaf Schenk. 2021. "New Frontiers in Bayesian Modeling Using the INLA Package in r." *Journal of Statistical Software* 100: 1–28.
- Vicente, G, T Goicoa, and MD Ugarte. 2020. "Bayesian Inference in Multivariate Spatio-Temporal Areal Models Using INLA: Analysis of Gender-Based Violence in Small Areas." *Stochastic Environmental Research and Risk Assessment* 34 (10): 1421–40.
- Villejo, Stephen Jun, Sara Martino, Finn Lindgren, and Janine B Illian. 2025. "A Data Fusion Model for Meteorological Data Using the INLA-SPDE Method." *Journal of the Royal Statistical Society Series C: Applied Statistics* 74 (4): 1021–56.